

# Generated Question Paper

**Q1.** The testes are a part of the human body. Meiosis is the process of dividing the testes into two parts. [3 marks]

One has been completed for you.

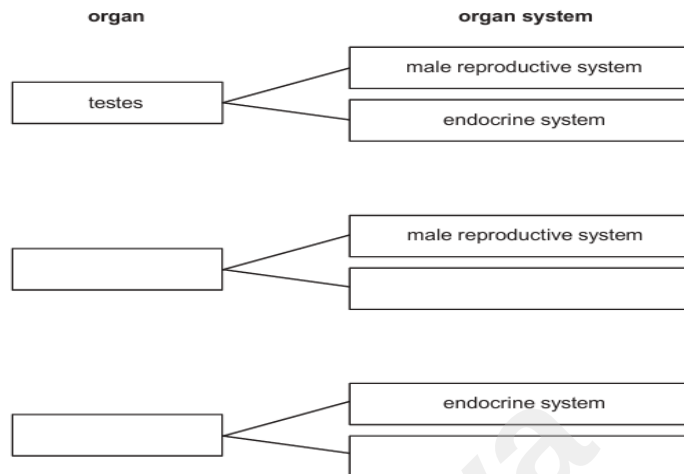


Fig. 5.1

Figure: organs\_two\_systems.png

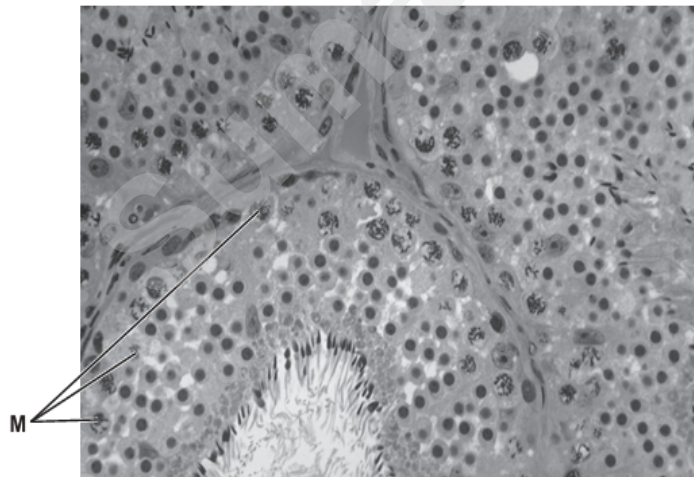


Fig. 5.2

Figure: mammalian\_testis\_micrograph.png

**Q2.** The effect of pH on lipase activity is shown on Fig.1. [6 marks]

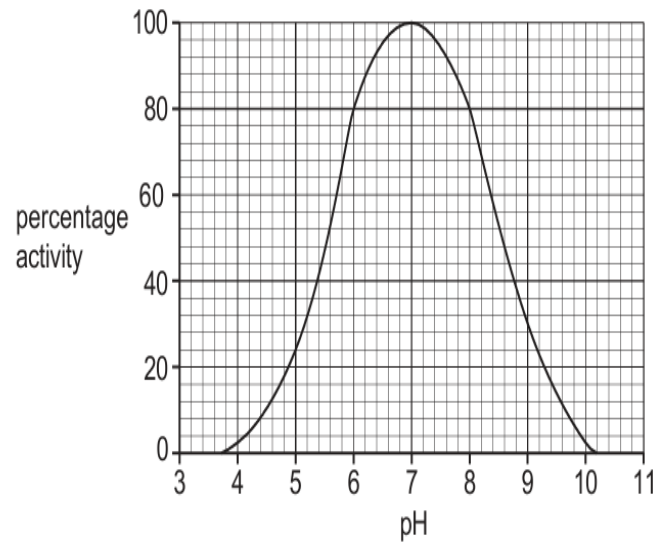


Figure: Fig1\_1\_LipaseActivity.png

**Q3.** Plants store starch in their leaves. [2 marks]

**Q4.** The process of converting ammonium ions into nitrate ions is called the nitrate reaction. [1 marks]

**Q5.** (c) Complete Table 2.1 by stating two structural features of epidermis and mesophyll tissues and explain how each feature is an adaptation for photosynthesis. [4 marks]

**Q6.** (ii) Name the process that causes gas bubbles to form in the dough. [1 marks]

**Q7.** (b) State the organs that make up the central nervous system. [1 marks]

**Q8.** The term 'species' is a term used to describe a species. [2 marks]



Fig. 6.1

Figure: potato\_varieties.png

**Q9.** (d) Explain why reflexes occur in unconscious people. [1 marks]

**Q10.** The advantage of using a pyramid of energy is that it is more efficient. [2 marks]

**Q11.** (e) shows a synapse and parts of two neurons. (e) begins center end center (i) Draw an arrow to show the direction of the signal across the synapse. [1 marks]

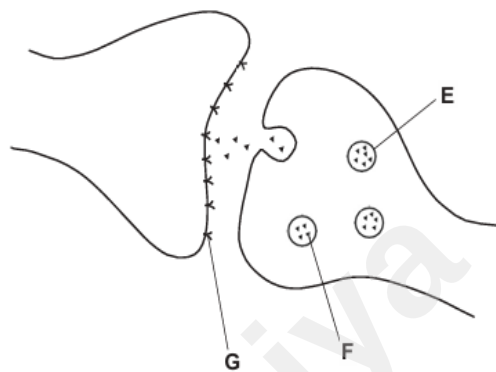


Fig. 4.1

Figure: synapse\_diagram.png

**Q12.** Enzymes are catalysts. (a) (i) Describe what is meant by the term catalyst. [2 marks]

**Q13.** The average photosynthesis rate of soybean plants was measured in two plots with different CO<sub>2</sub> concentrations, 370 ppm and 550 ppm. Describe and explain the effect of carbon dioxide concentration on the average rates of photosynthesis of the soybean plants from 04:00 to 22:00. [6 marks]

The atmospheric carbon dioxide concentrations in the two plots were kept at:

- 370 ppm, which is similar to the current atmospheric carbon dioxide concentration
- 550 ppm, which is a possible future atmospheric carbon dioxide concentration.

When the soybean plants were fully grown, scientists calculated the average rates of photosynthesis at regular intervals from 04:00 to 22:00 for both plots.

The results are shown in Fig. 2.1.

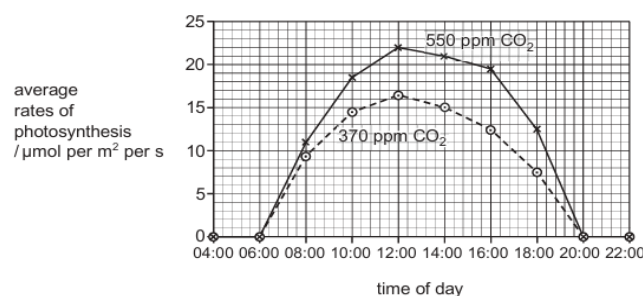


Fig. 2.1

Figure: photosynthesis\_rate\_graph.png

**Q14.** A: Blood vessel C. [2 marks]

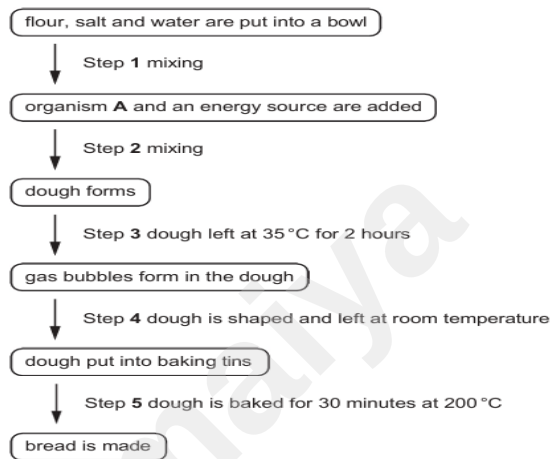
**Q15.** The retina is a part of the retina. What is the retina's function? [3 marks]

**Q16.** (b) (i) Name the source of energy used by organism A. [1 marks]

**Q17.** The name of organism A is a comma. [1 marks]

Biotechnology is used in the process of bread-making.

Fig. 1.1 shows some of the steps in making bread.



**Fig. 1.1**

Figure: bread\_making\_steps.png

**Q18.** The nitrogen cycle is a process that occurs in the nitrogen cycle. [1 marks]